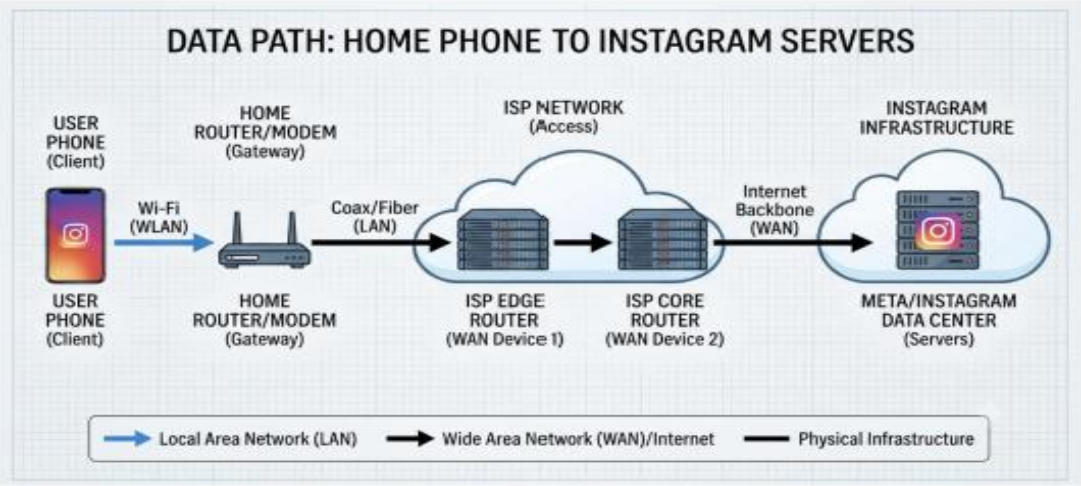
Wide Area Networks

1.Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.



2.Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

WAN Technology	Speed	Cost	Real-Life Use
MPLS	High and reliable	High	Used by large companies like Zomato to securely connect offices in different cities.
Leased Line	High and dedicated	Very high	Used by banks and IRCTC for reliable connectivity and important online services.
Broadband Internet	Medium to high	Low	Used in homes, small offices, and shops for browsing, online classes, and streaming.

**3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

Hint: Focus on how the VPN creates a private tunnel over the public WAN.**

A VPN (Virtual Private Network) allows users to access the internet securely through a WAN (Wide Area Network), such as the public internet. It creates a private, encrypted tunnel between the user's device and a VPN server, protecting data while it travels across the public network.

For example, when I visit a coffee shop and connect my laptop to its public Wi-Fi, I may want to use apps like Spotify or YouTube. Public Wi-Fi can be risky because attackers on the same network may try to intercept unprotected data.

A VPN helps protect my connection in the following ways:

1. Connection to the VPN: I connect my laptop to the coffee shop's public Wi-Fi and turn on the VPN application.

2. Encrypted tunnel: The VPN encrypts my internet traffic and creates a secure tunnel through the public WAN. This makes the data difficult for unauthorized people to read.

3. Secure data transmission: My encrypted data travels through the internet to the VPN server, where it is decrypted and forwarded to the required service, such as YouTube or Spotify.

4. IP address protection: Websites and apps generally see the VPN server's IP address instead of my actual public IP address, helping protect my online privacy.

5. Receiving data securely: When YouTube or Spotify sends information back, the response travels through the VPN connection to my device, where the VPN decrypts it.

Example: If I watch a YouTube video using public Wi-Fi at a coffee shop, the VPN helps protect my internet traffic from people trying to spy on the local Wi-Fi connection.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Recommended WAN Technology: MPLS (Multiprotocol Label Switching)

For Flipkart, I would recommend MPLS to connect its warehouses in Ahmedabad, Mumbai, and Delhi.

Reasons:

1. Reliable and Fast Connection: MPLS provides reliable, high-speed connectivity between warehouses, helping Flipkart manage inventory, track orders, and update delivery information efficiently.

2. Secure Communication: MPLS supports private network connectivity between warehouse locations, helping protect business data and ensuring smooth communication between different branches.